

Minh Duong

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EDUCATION

MiraCosta College Technology Career Institute

09/2025 - 12/2025

- 600+ hours of hands-on training in robotics, electronics testing, soldering, wiring, and system-level troubleshooting.

University of California, San Diego

09/2022 - 06/2024

Bachelor of Science in Computer Science

EXPERIENCE

Engineering Technician | *SMAC*

02/2026 - Present

- Performed electrical testing using **HiPot testers and oscilloscopes** to validate unit safety, signal integrity, and performance.
- Executed automated and manual test programs to identify functional defects and ensure compliance with specifications.
- Documented detailed **test reports** including measurements, failure analysis, and validation results for each unit.
- Modified and optimized test programs to support **engineering changes and unit variations**.
- Troubleshoot and isolated **root-cause failures** in electromechanical systems, improving reliability and reducing rework.
- Designed **PCB layouts and flex cables** using *Autodesk Eagle* based on system requirements.
- Developed **3D mechanical models** using *SolidWorks* to support design validation and integration.
- Performed **precision soldering and rework** on connectors and PCB components per **IPC-620 wiring standards**.
- Replaced and repaired **faulty electrical components** on circuit boards to restore functionality and meet quality requirements.
- Supported engineering team by performing PCB rework and modifications, reducing turnaround time for updated units.
- Conducted pneumatic testing using vacuum and pressure cycles to verify threshold requirements.
- Performed leak testing by submerging units in water to detect air leaks and ensure system integrity.
- Interpreted electrical schematics to understand board-level connections and support testing, troubleshooting, and rework.
- Collaborated across software, electrical, and design engineering teams, as well as assembly and fabrication lines, to support unit development and testing.

Technical Support Engineer Internship | *XILO*

02/2025 – 03/2025

- Developed the XAI Extension to streamline insurance quoting, reducing manual data entry by **75%** and increasing agent efficiency.
- Supported technical operations by following documented procedures to **configure, test, and validate system functionality**.
- Debugged and resolved system issues by analyzing **data flow**, identifying **root causes**, and documenting **corrective actions**.
- Collaborated cross-functionally with engineers and stakeholders to implement updates, verify results, and support reliable system operation in a fast-paced environment.

PROJECTS

Smart Home Automation System | *Arduino, Fusion 360, GitHub, Sensors, Laser Cut*

12/2025

- Fabricated, assembled, installed, and modified a multi-subsystem electromechanical system supporting mechanical structures, wiring, sensors, motors, and controllers.
- Set up and checked out test configurations following pre-defined procedures and documented results.
- Implemented and tested a secure access gate using PIR sensing, RFID authentication, and servo actuation.
- Built and validated an automated clothesline using weather sensing and motor-driven pulley and gearbox assemblies.
- Performed system-level testing, calibration, and troubleshooting to verify mechanical, electrical, and sensor performance.
- Supported prototype test setup, instrumentation, and measurement recording during system evaluations.

Mouse Maze | *Arduino, Ultrasonic Sensor, Servo Motor, Fusion*

10/2025

- Developed an autonomous robotic chassis with obstacle detection and avoidance using ultrasonic sensors and servo control.
- Designed and fabricated 3D-printed components in Fusion 360 to protect electronics and maintain structural alignment.
- Integrated mechanical and electrical subsystems; optimized system performance through testing, calibration, and design adjustments.

SKILLS & CERTIFICATIONS

- Testing & Troubleshooting:** Oscilloscope, multimeter, power supply, HiPot testing, signal verification, root cause analysis.
- Electrical & Mechanical Systems:** Circuits, motors, servos, actuators, sensors (ultrasonic, PIR, RFID), pneumatics.
- Automation & Controls:** PLC fundamentals, I/O systems, ladder logic basics, system commissioning.
- Design & Tools:** SolidWorks, Fusion 360, Autodesk Eagle, 3D printing, hand tools, caliper, laser-cut parts, Dremel.
- Software:** Python, C++, VS Code, Arduino IDE, MySQL, Git, Arduino IDE, MS Office, Jira, eDrawings.
- Systems:** Windows, Linux (Ubuntu), Unix.
- Certifications:** OSHA-30, Lean Yellow Belt, 5S, Equipment Technician.